UFUG 2104
Written Assignment 1

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Tips: Please clilck the link below to get to the targeted answers.

Problem 1
Problem 2
Problem 3
Problem 4
Problem 5

1 Problem 1

(16 pts) The pdf (probability density function) of X is given by

$$f_X(x) = \begin{cases} -\frac{1}{4}x, & -2 \le x < 0 \\ cx^2, & 0 \le x \le 3 \\ 0, & \text{otherwise} \end{cases}$$

(a) Find the value of the constant c.

(b) Plot the pdf of X from -3 to 4. If you plot by hand, make sure that your plot shows the main features, i.e., where they intercept the y-axis and x-axis. (c) Find the expression for the cdf (cumulative distribution function) of X. (d) Find the expected value of X. (e) Find the probability of the event $A = \{1 \le x \le 2\}$. (f) Find the conditional pdf of X given A, $f_{X|A}(x|A)$ and plot it from -3 to 4. If you plot by hand, make sure that your plot shows the main features, i.e. where they intercept the y- axis and x-axis. (g) Find the conditional expected value of X given A, $\mathbb{E}[X|A]$

Solution

(a) By the definition of pdf, we have

$$\int_{-\infty}^{\infty} f_X(x)dx = 1$$

and given the property of the function, we have

$$\int_{-\infty}^{\infty} f_X(x)dx = \int_{-\infty}^{-2} 0dx + \int_{-2}^{0} \left(-\frac{1}{4}x\right) dx + \int_{0}^{3} (cx^2)dx + \int_{4}^{\infty} 0dx = 1$$

Which means

$$(-\frac{1}{8}x^2)\Big|_{-2}^{0} + (\frac{1}{3}cx^3)\Big|_{0}^{3} = \frac{1}{2} + 9c = 1$$

$\Rightarrow \boxed{c = \frac{1}{18}}$

(b) The pdf can be graphed as below:

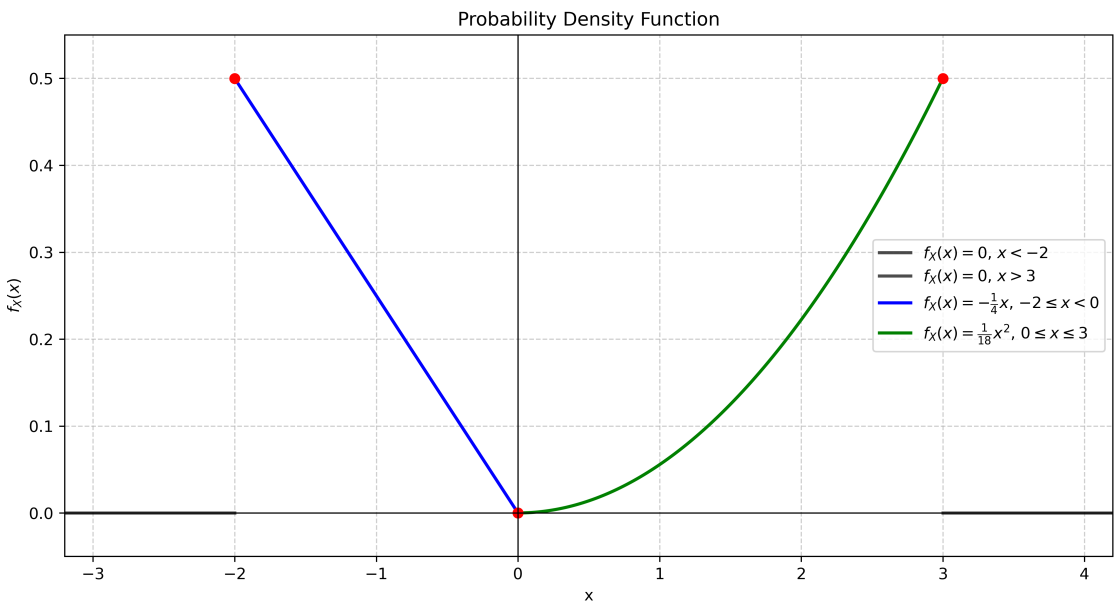

Figure 1

(c) The cdf can be calculated as below:

$$\boxed{F_X(x) = \begin{cases} 0 & x < -2 \\ \frac{1}{2} - \frac{1}{8}x^2 & -2 \le x < 0 \\ \frac{1}{2} + \frac{1}{54}x^3 & 0 \le x \le 3 \\ 1 & 3 < x \end{cases}}$$

(d) the expected value can be cal culated as below: when x < -2, the expected value is 0; when -2 <= x < 0, the expected value is

$$\int_{-2}^{0} -x(\frac{1}{4}x)dx = -\left(\frac{1}{12}x^3\right)\Big|_{x=-2}^{0} = -\frac{2}{3}$$

when $0 \le x \le 3$, the expectation is

$$\int_{0}^{3} x\frac{1}{18}x^2 dx = \left(\frac{1}{72}x^4\right)\Big|_{x=0}^{3} = \frac{9}{8}$$

And the same expectation value of 0m can be applied to the case when $x > 3$. So the total expectation for the expectation of

$$\boxed{\mathbb{E}(X) = -\frac{2}{3} + \frac{9}{8} = \frac{11}{24}}$$

(e) For the event $A = \{1 \le x \le 2\}$, the probability can be calculated in this way:

$$\boxed{\mathbb{P}(A) = \int_{1}^{2} \frac{1}{18}x^2 = \frac{1}{54}x^3\Big|_{x=1}^{2} = \frac{7}{54}}$$

(f) When $x \in [1, 2]$, the conditional pdf can be calculated as below:

$$f_{X|A}(x|A) = \frac{f_X(x)}{\mathbb{P}(A)} = \frac{\frac{1}{18}x^2}{\frac{7}{54}} = \frac{3}{7}x^2, \quad x \in [1, 2]$$

When $x \notin [1, 2]$, the conditional pdf can be calculated as 0. So we have:

$$\boxed{f_{X|A}(x|A) = \begin{cases} \frac{3}{7}x^2 & x \in [1, 2] \\ 0 & \text{otherwise} \end{cases}}$$

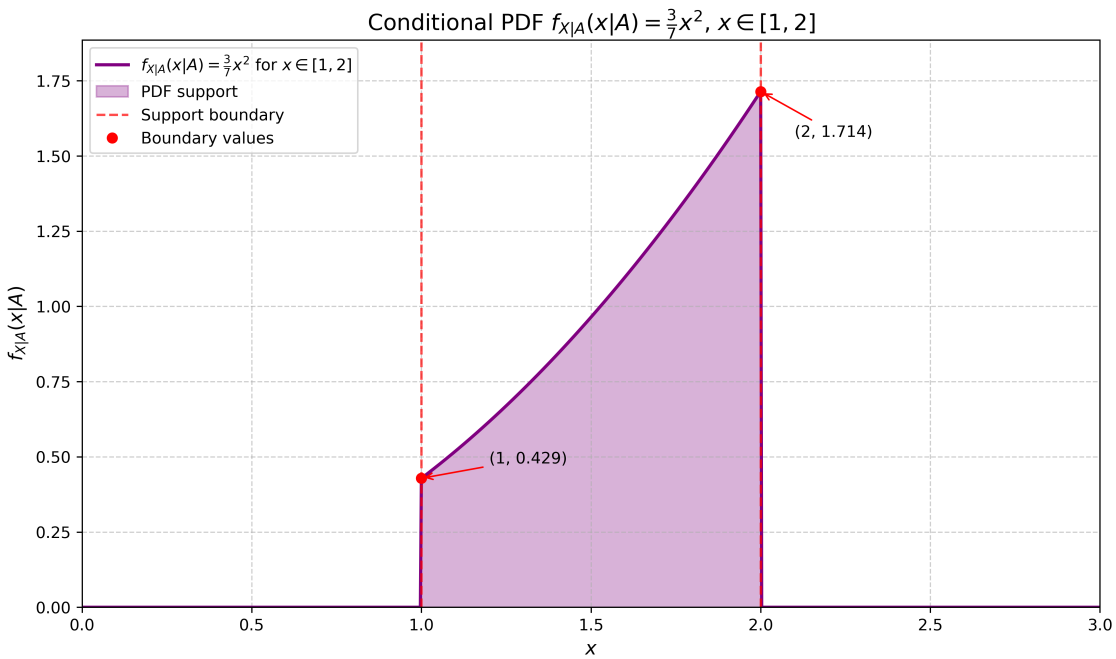

Figure 2

(g) we only have to calculate the conditional Expectation when $x \in [1, 2]$:

$$\boxed{\mathbb{E}[X|A] = \int_1^2 \frac{3}{7}x^2 = \frac{9}{7}}$$

2 Problem 2

(16 pts) Suppose we have two machines. The lifetime of Machine 1 in years, X_1, follows an exponential distribution with parameter $\lambda_1 = \frac{1}{9}$. The lifetime of Machine 2 in years, X_2, follows an exponential distribution with parameter $\lambda_2 = \frac{1}{5}$.

(a) Compute the expected values of X_1 and X_2. Which machine lasts longer on average?

(b) Find the conditional pdf of X_2 given that the lifetime of Machine 2 is less than 4 years.

(c) Compute the conditional expected value of X_2 given that the lifetime of Machine 2 is less than 4 years.

(d) Find the probability that the lifetime of Machine 1 exceeds 9 years. Repeat for Machine 2.

(e) Find the probability that Machine 1 will continue to work for at least another 6 years, given that the machine has been working for 9 years already

Solution

(a)

Lemma We have the expectation value:

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} \lambda e^{-\lambda x} dx = \int_{-\infty}^{0} \lambda e^{-\lambda x} dx + \int_{0}^{\infty} \lambda e^{-\lambda x} dx = 0 + (-e^{-\lambda x})\Big|_0^{\infty} = \frac{1}{\lambda}$$

So the expectation of the Random Variables X_1, X_2 are:

$$\boxed{\mathbb{E}[X_1] = \frac{1}{\lambda_1} = 9, \quad \mathbb{E}[X_2] = \frac{1}{\lambda_2} = 5}$$

(b) By the reality meaning of the exponential distribution of the lifetime of the machine, we have $x \geq 0$ And the probability of $0 < x < 4$, we have

$$\mathbb{P}_X[0 < x < 4] = \int_0^4 \frac{e^{-\frac{x}{5}}}{5} = -e^{-\frac{x}{5}}\Big|_0^4 = 1 - e^{-\frac{4}{5}}$$

And the condotional pdf can be divided to 2 situations: when $0 < x < 4$, we have

$$\mathbb{P}_X[0 < x < x_1] = \int_0^x \frac{e^{-\frac{x_1}{5}}}{5} = 1 - e^{-\frac{x_1}{5}}$$

So the conditional pdf can be written as:

$$\mathbb{P}[X] = \frac{\frac{1}{5}e^{-\frac{x_1}{5}}}{1 - e^{-\frac{4}{5}}}$$

when $x > 4$, and we have:

$$\mathbb{P}[X] = 1$$

So in conclusion, we have

$$\mathbb{P}[X] = \begin{cases} \frac{\frac{1}{5}e^{-\frac{x_1}{5}}}{1-e^{-\frac{4}{5}}} & 0 \leq x < 4 \\ 0 & 4 < x \end{cases}$$

we will have the conditional pdf as below:

(c) We will have the conditional expected value of X_2 given $X_2 < 4$ By definition,

$$\mathbb{E}[X_2|X_2 < 4] = \int_0^\infty x f_{X_2|X_2<4}(x)dx = \frac{1}{5(1-e^{-\frac{4}{5}})}\int_0^4 xe^{-\frac{x}{5}}dx = 25 - 45e^{-\frac{4}{5}}$$

And we can calculate the numerical value

$$\boxed{\mathbb{E}[X_2|X_2 < 4] = \frac{5-9e^{-\frac{4}{5}}}{1-e^{-\frac{4}{5}}}} \approx 1.734\text{years}$$

(d) for the exponential distribution,

$$\mathbb{P}[X > t] = e^{-\lambda t}$$

for machine 1,

$$\mathbb{P}[X_1 > 9] = e^{-1}$$

for machine 2,

$$\mathbb{P}[X_2 > 9] = e^{\frac{9}{5}}$$

So

$$\boxed{\mathbb{P}[X_1 > 9] = e^{-1} \approx 0.3679, \quad \mathbb{P}[X_2 > 9] = e^{-\frac{9}{5}} \approx 0.1653.}$$

(e) We want

$$\mathbb{P}[X_1 > 15|X_1 > 9] = \frac{\mathbb{P}[X_1 > 15]}{\mathbb{P}[X_1 > 9]}$$

Since $X_1 \propto E\left[\frac{1}{9}\right]$

$$\mathbb{P}(X_1 > 15) = e^{-\frac{5}{3}}, \quad \mathbb{P}(X_1 > 9) = e^{-1}$$

Hence

$$\boxed{\mathbb{P}(X_1 > 15|X_1 > 9) = e^{-\frac{2}{3}}}$$

3 Problem 3

The joint pdf of X and Y is given by:

$$f_{X,Y}(x,y) = \begin{cases} c(x+y)e^{-y}, & -y \le x \le y, \quad 0 \le y \le \infty \\ 0, & \text{otherwise} \end{cases}$$

For this question, it may be useful to draw the region in the X, Y plane where the pdf is non- zero to help you determine the limits of the integrals.

(a) Find the value of the constant c.

(b) Find the marginal pdfs of X and Y, respectively.

(c) Find the probability that $X > 2$ and $Y > 2$, $P[X > 2 \cap Y > 2]$.

(d) Find the conditional pdf of X given Y, $f_{X|Y}(x|y)$. Plot $f_{X|Y}(x|y)$ for $y = 2$.

(e) Find the conditional expectation of X given $Y = y$, $\mathbb{E}[X|y]$. Plot $\mathbb{E}[X|y]$ for y from 0 to 6.

Solution

(a) We have the integration of the pdf must be 1, so we can integrate the pdf to get the c

$$\int_0^\infty \int_{-y}^{y} c(x+y)e^{-y}dxdy = 1$$

we integrate the x first:

$$\int_{-y}^{y} (x+y)dx = \int_{-y}^{y} xdx + \int_{-y}^{y} ydx = 0 + 2y^2 = 2y^2$$

So the

$$\int_{-y}^{y} = c(x+y)e^{-y}dx = 2cy^2e^{-y}$$

And substitude it into the original equation,

$$\int_0^\infty 2cy^2e^{-y}dy = 1$$

By the Gamma function identity, we have

$$\int_0^\infty y^2e^{-y}dy = 2! = 2$$

So we have

$$\boxed{c = \frac{1}{4}}$$

(b) Consider the marginal of Y,

$$f_Y(y) = \int_{-y}^{y} f_{X,Y}(x,y)dxdy$$

Substituting the $c = \frac{1}{4}$

$$f_Y(y) = \frac{1}{4}e^{-y}\int_{-y} y(x+y)dx = \frac{1}{2}y^2e^{-y}, \quad y \geq 0$$

So the marginal of Y,

$$\boxed{f_Y(y) = \begin{cases} \frac{1}{2}y^2e^{-y} & y \geq 0 \\ 0 & y < 0 \end{cases}}$$

And consider the m,arginal of X, given that $|x| < y$ so

$$f_X(x) = \int_{|x|}^{\infty} f_{X,Y}(x,y)dy$$

Substituting the $c = \frac{1}{4}$

$$f_X(x) = \int_{|x|}^{\infty} \frac{1}{4}(x+y)e^{-y}dy = \frac{1}{4}\left(x\int_{|x|}^{\infty} e^{-y}dy + \int_{|x|}^{\infty} ye^{-y}dy\right)$$

if $x > 0$, then

$$f_X(x) = \frac{1}{4}(2x+1)e^{-x}$$

if $x < 0$, then

$$f_X(x) = \frac{1}{4}\left(xe^x + (1-x)e^x\right) = \frac{1}{4}e^x$$

So the equationwe have

$$\boxed{f_X(x) = \begin{cases} \frac{1}{4}e^x & x < 0 \\ \frac{1}{4}(2x+1)e^x & x \geq 0 \end{cases}}$$

(c)

$$p(X > 2, Y > 2) = \int_2^{\infty} \int_2^{y} \frac{1}{4}(x+y)e^{-y} dx dy$$

integrate the X first,

$$\int_2^{y} (x+y)dx = \left[\frac{x^2}{2} + yx\right]_2^{y} = \frac{3}{2}y^2 - 2y - 2$$

And substitude this formula,

$$\frac{1}{4}\int_2^{\infty} \left(\frac{3}{2}y^2 - 2y - 2e^{-y} dyb\right) = \frac{1}{4}\left(\frac{3}{2}10e^{-2} - 2 \cdot 3e^{-2} - 2e^{-2}\right) = \boxed{\frac{7}{4}e^{-2}}$$

(d) given that

$$f_{X,Y}(x,y) = \frac{1}{4}(x+y)e^{-y}, \quad f_Y(y) = \frac{1}{2}e^{-}y \quad y \geq 0$$

So

$$f_{X|Y}(x|y) = \begin{cases} \frac{x+y}{2y^2} & x \leq |y|,\ y > 0 \\ 0 & \text{otherwise} \end{cases}$$

For $y = 2$

$$f_{X|Y}(x|2) = \begin{cases} \frac{x+2}{8} & -2 \leq x \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

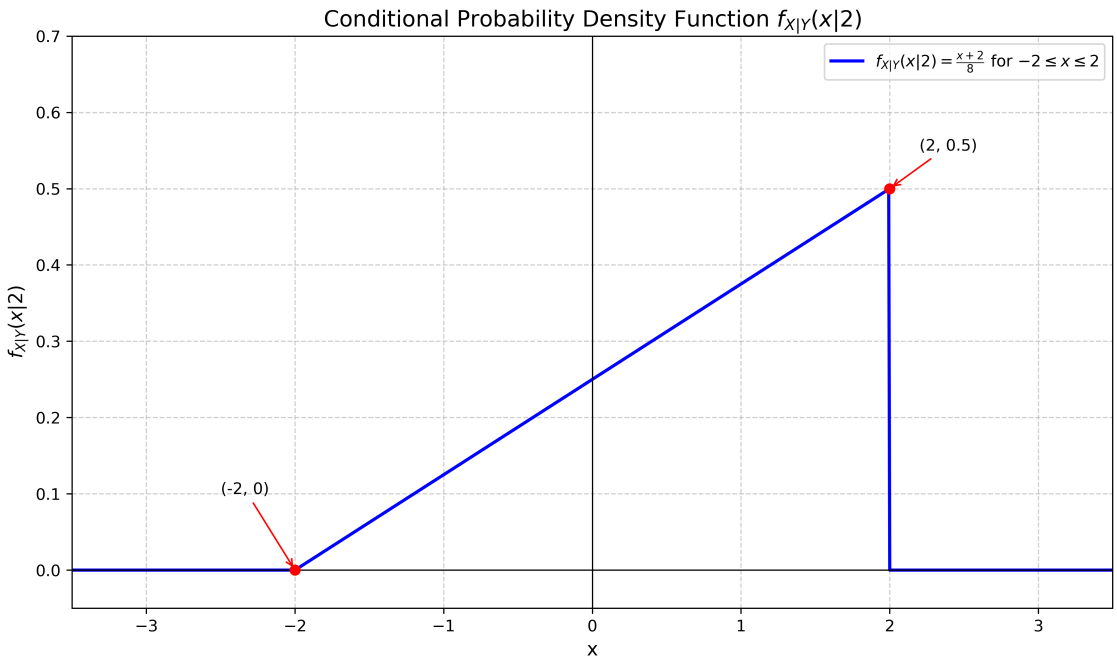

Figure 3

(e) By the conditional density,

$$\mathbb{E}[X|Y=y] = \int_{-y}^{y} x f_{X|Y}(x|y)dx = \int_{-y}^{y} x\frac{x+y}{2y^2}dx = \frac{1}{2y^2}\int_{-y}^{y}(x^2+xy)dx = \frac{1}{2y^2}\left(\frac{2}{3}y^3\right) = \frac{1}{3}y, \quad y \geq 0$$

$$\boxed{\mathbb{E}[X|Y=y] = \frac{1}{3}y}$$

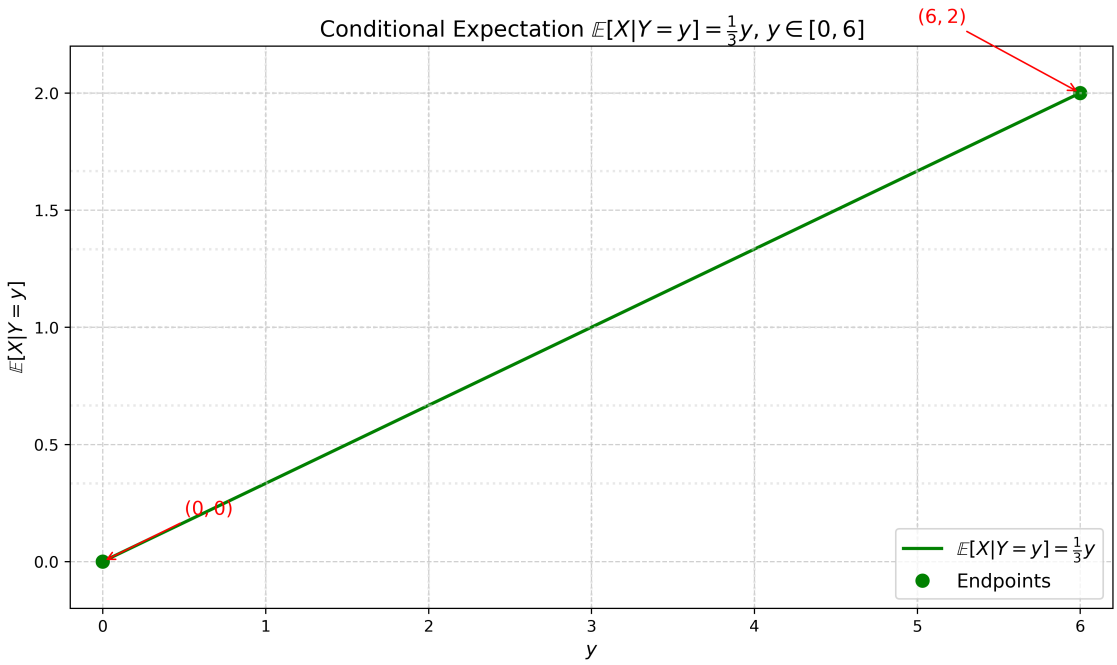

Figure 4

4 Problem 4

(12 pts) Let X and Y be jointly Gaussian random variables with means $\mathbb{E}[X] = 1$ and $E[Y] = -4$, variances $\text{Var}(X) = 4$ and $VarY = 9$. Define a random variable $Z = -2X + Y$.

(a) Assume that X and Y are uncorrelated. Find the pdf of Z, $f_Z(z)$ for $-\infty < z < \infty$.

(b) Assume that X and Y have a correlation coefficient $\rho = -0.3$. Find the pdf of Z, $f_Z(z)$ for $-\infty < z < \infty$.

Solution

(a) Since X and Y are jointly Gaussian, any linear combination of X and Y is also Gaussian. Hence,

$$Z = -2X + Y$$

is Gaussian. Therefore, to determine the distribution of Z, it suffices to compute its mean and variance.

First, by linearity of expectation,

$$\mathbb{E}[Z] = \mathbb{E}[-2X + Y] = -2\mathbb{E}[X] + \mathbb{E}[Y].$$

Substituting the given values,

$$\mathbb{E}[Z] = -2(1) + (-4) = -6.$$

Since X and Y are uncorrelated, we have

$$\text{Cov}\left(()\, X, Y\right) = 0.$$

Using the variance formula for a linear combination,

$$\text{Var}\left(()\, aX + bY\right) = a^2\text{Var}\left(()\, X\right) + b^2\text{Var}\left(()\, Y\right) + 2ab\,\text{Cov}\left(()\, X, Y\right),$$

we obtain

$$\text{Var}\left(()\, Z\right) = \text{Var}\left(()\, -2X + Y\right) = (-2)^2\text{Var}\left(()\, X\right) + 1^2\text{Var}\left(()\, Y\right) + 2(-2)(1)\text{Cov}\left(()\, X, Y\right).$$

Substituting the given values,

$$\text{Var}\left(()\, Z\right) = 4 \cdot 4 + 1 \cdot 9 + 2(-2)(1)(0) = 16 + 9 = 25.$$

Thus,

$$Z \sim N(-6, 25).$$

Therefore, the pdf of Z is

$$f_Z(z) = \frac{1}{\sqrt{2\pi(25)}} \exp\left(-\frac{(z-(-6))^2}{2(25)}\right) = \frac{1}{5\sqrt{2\pi}} \exp\left(-\frac{(z+6)^2}{50}\right), \qquad -\infty < z < \infty.$$

(b) Since X and Y are jointly Gaussian, the random variable

$$Z = -2X + Y$$

is also Gaussian. Thus, we again compute its mean and variance.

By linearity of expectation,

$$\mathbb{E}[Z] = -2\mathbb{E}[X] + \mathbb{E}[Y] = -2(1) + (-4) = -6.$$

Next, we use the relation between covariance and correlation:

$$\text{Cov}\left(()\, X, Y\right) = \rho\, \sigma_X \sigma_Y,$$

where

$$\sigma_X = \sqrt{\text{Var}\left(()\, X\right)} = \sqrt{4} = 2, \qquad \sigma_Y = \sqrt{\text{Var}\left(()\, Y\right)} = \sqrt{9} = 3.$$

Since $\rho = -0.3$, we get

$$\text{Cov}\left(()\, X, Y\right) = (-0.3)(2)(3) = -1.8.$$

Now apply the variance formula for a linear combination:

$$\text{Var}\left(()\, Z\right) = \text{Var}\left(() - 2X + Y\right) = (-2)^2\text{Var}\left(()\, X\right) + 1^2\text{Var}\left(()\, Y\right) + 2(-2)(1)\text{Cov}\left(()\, X, Y\right).$$

Substituting the known values,

$$\text{Var}\left(()\, Z\right) = 4 \cdot 4 + 1 \cdot 9 + 2(-2)(1)(-1.8) = 16 + 9 + 7.2 = 32.2.$$

Hence,

$$Z \sim N(-6, 32.2).$$

Therefore, the pdf of Z is

$$f_Z(z) = \frac{1}{\sqrt{2\pi(32.2)}} \exp\left(-\frac{(z-(-6))^2}{2(32.2)}\right) = \frac{1}{\sqrt{64.4\pi}} \exp\left(-\frac{(z+6)^2}{64.4}\right), \qquad -\infty < z < \infty.$$

5 Problem 5

(8 pts) Assume that the monthly salaries of Hong Kong residents are i.i.d random variables with unknown mean, but assume the standard deviation is 15,000HKD. Alice wants to use the sample mean to estimate the mean value of the monthly salaries of Hong Kong residents.

(a) Use the Central Limit Theorem to approximate the number of Hong Kong residents that should be surveyed, n, such that the sample mean is within 1000HKD of the actual mean with probability greater than or equal to 96%.

(b) Suppose that the true mean monthly salary of a Hong Kong resident is 18,000HKD. Suppose we average the monthly salary of 500 Hong Kong residents. What is the probability that the measured average is within 1000HKD of the true mean?

Solution

(a) Let $X_1, X_2, \ldots, X_n$ denote the monthly salaries of sampled Hong Kong residents, where the variables are i.i.d. with mean μ and standard deviation

$$\sigma = 15000.$$

Let $\overline{X}$ be the sample mean. By the Central Limit Theorem, for sufficiently large n,

$$\overline{X} \approx N\left(\mu, \frac{\sigma^2}{n}\right) = N\left(\mu, \frac{15000^2}{n}\right).$$

We want

$$P\left(|\overline{X} - \mu| \leq 1000\right) \geq 0.96.$$

Standardizing,

$$P\left(|\overline{X} - \mu| \leq 1000\right) = P\left(\left|\frac{\overline{X} - \mu}{15000/\sqrt{n}}\right| \leq \frac{1000}{15000/\sqrt{n}}\right) = P\left(|Z| \leq \frac{\sqrt{n}}{15}\right),$$

where $Z \sim N(0, 1)$.

Since

$$P(|Z| \leq z) = 0.96,$$

we need

$$z = z_{0.98} \approx 2.054,$$

because the two tails together have probability 0.04, so each tail has probability 0.02.

Thus,

$$\frac{\sqrt{n}}{15} \geq 2.054.$$

Hence,

$$\sqrt{n} \geq 15(2.054) = 30.81,$$

so

$$n \geq (30.81)^2 \approx 949.26.$$

Therefore, the minimum required sample size is

$$\boxed{n = 950.}$$

(b) Now suppose that the true mean monthly salary is

$$\mu = 18000,$$

and the sample size is $n = 500$. By the Central Limit Theorem,

$$\overline{X} \approx N\left(18000, \frac{15000^2}{500}\right).$$

We want to compute

$$P\left(|\overline{X} - 18000| \leq 1000\right).$$

Standardizing,

$$P\left(|\overline{X} - 18000| \leq 1000\right) = P\left(\left|\frac{\overline{X} - 18000}{15000/\sqrt{500}}\right| \leq \frac{1000}{15000/\sqrt{500}}\right).$$

Thus,

$$P\left(|\overline{X} - 18000| \leq 1000\right) = P\left(|Z| \leq \frac{1000\sqrt{500}}{15000}\right) = P\left(|Z| \leq \frac{\sqrt{500}}{15}\right),$$

where $Z \sim N(0, 1)$.

Now,

$$\frac{\sqrt{500}}{15} \approx 1.491.$$

Therefore,

$$P\left(|Z| \leq 1.491\right) = 2\Phi(1.491) - 1.$$

Using the standard normal table,

$$\Phi(1.491) \approx 0.932.$$

Hence,

$$P\left(|\overline{X} - 18000| \leq 1000\right) \approx 2(0.932) - 1 = 0.864.$$

Therefore, the required probability is approximately

$$\boxed{0.864.}$$